

What happened during the COVID-19 pandemic?

ME317: Statistical Methods for Risk Management. Project 2026

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1. Introduction

The COVID-19 pandemic caused one of the fastest stock-market crashes in history, followed by an unusual recovery. This report uses the methods from ME317 to study this period in four steps. Firstly, it builds a clean data set of daily prices for 12 US stocks from six sectors over 2019–2021. Secondly, it studies the returns, identifies the extreme days, links them to news events, and tests whether a normal distribution can describe the data. Thirdly, it forms a buy-and-hold portfolio, estimates its Value-at-Risk (VaR) and Expected Shortfall (ES) with three models, and backtests them on an out-of-sample year. Finally, it studies the dependence between the stocks with copulas and tests whether the selected copula can reproduce simultaneous crashes. The full R code is in the appendix, and every numerical result is generated by that code.

2. Data

Source and structure. Daily prices were downloaded from Yahoo Finance with the R package `quantmod`, using `getSymbols(..., src = "yahoo")`. The data set contains daily *adjusted* closing prices for 12 large US stocks from six sectors: financials (JPM, GS), energy (XOM, CVX), technology (AAPL, MSFT), consumer (PG, MCD), healthcare (PFE, JNJ) and industrials/airlines (BA, DAL). The sample runs from 02 January 2019 to 31 December 2021 and contains 757 trading days. From these prices, 756 daily log returns are calculated as $r_t = \ln(P_t/P_{t-1})$.

Cleaning. Adjusted prices are used because Yahoo Finance corrects them for stock splits and dividend payments; raw closing prices would show artificial jumps on those dates. The merged data set contains no missing values, so no imputation is needed; the code nevertheless applies last-observation-carried-forward and removes incomplete leading rows as a safeguard if gaps occur. The data are also checked for one-day price changes greater than 50% in absolute value, which could indicate a data error rather than a real market movement (no such observations are found).

Sample split. The years 2019–2020 contain 504 observations, including the main COVID-19 crash, and are used to estimate all models. The 252 observations from 2021 are kept as an out-of-sample period for the backtests in Section 4.

3. Returns, Extreme Events, and Non-Normality

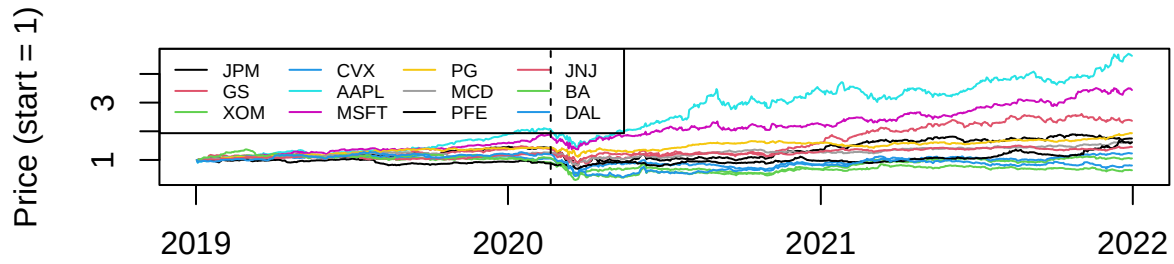


Figure 1: Normalised stock prices of the 12 companies, 2019–2021 (the dashed line marks 20 February 2020, the start of the COVID-19 crash).

Figure 1 shows the normalised price paths. All 12 stocks fell together in late February and March 2020, but their recoveries were very different: technology stocks reached new highs within months, while airline and energy stocks remained weak for much longer. This is the first sign that both the size of price movements and the co-movement between stocks changed during the crisis.

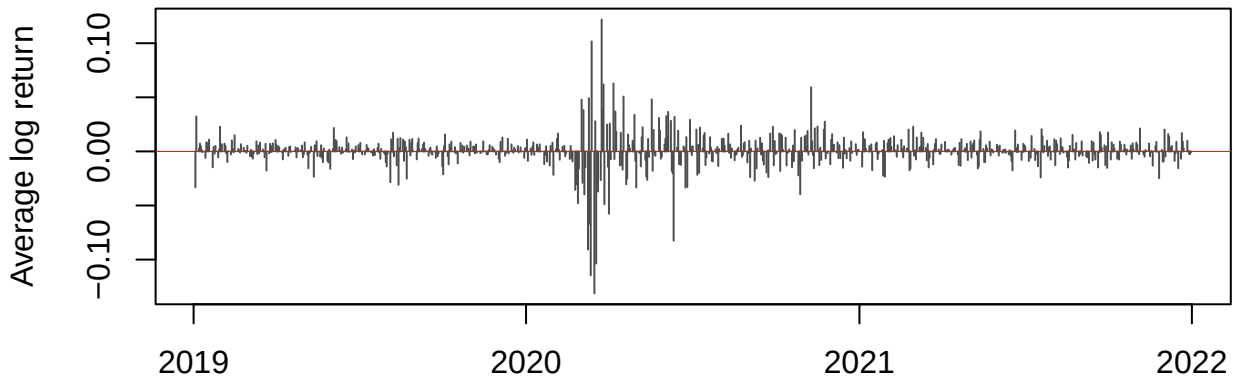


Figure 2: Daily log-return dynamics: equal-weighted average return of the 12 stocks.

Figure 2 shows two typical features of financial returns. Firstly, volatility is clustered: the market is quiet in 2019, highly volatile in March 2020, and calmer again in 2021 — volatility comes in waves instead of staying constant. Secondly, the extreme days are much larger than most other observations, which already suggests that a thin-tailed model may not describe these returns well.

Table 1: Five worst and five best days, measured by the equal-weighted average log return of the 12 stocks.

Date	Avg log return (%)	Type
16 Mar 2020	-13.12	Worst
12 Mar 2020	-11.46	Worst
18 Mar 2020	-10.36	Worst
09 Mar 2020	-9.07	Worst
11 Jun 2020	-8.25	Worst
24 Mar 2020	12.19	Best
13 Mar 2020	10.17	Best
06 Apr 2020	6.27	Best
26 Mar 2020	6.20	Best
09 Nov 2020	5.92	Best

Extreme events. Table 1 lists the five worst and five best days in the sample. Almost all extreme days occurred in March 2020 and can be linked to major news. On 9 March, the first market-wide circuit breaker was triggered during the oil-price conflict and the spread of the virus in Europe. The fall on 12 March followed the WHO pandemic declaration and the announcement of US travel restrictions from Europe. On 16 March, markets fell again after an emergency Federal Reserve rate cut to zero did not calm investors. The large positive days are mainly rebounds inside the same panic (13 and 24 March; the 24 March rise followed news of a \$2 trillion US fiscal package). On 9 November 2020, Pfizer and BioNTech announced that their vaccine was more than 90% effective; reopening stocks such as Delta Air Lines rose strongly, while many lockdown winners increased less.

Table 2: Summary statistics of daily log returns, 2019–2021 (a normal distribution has skewness 0 and kurtosis 3).

	Sector	Ann. mean	Ann. vol.	Skewness	Kurtosis	JB p-value
JPM	Financials	0.19	0.35	-0.09	16.5	<2e-16
GS	Financials	0.29	0.36	-0.15	13.4	<2e-16
XOM	Energy	0.02	0.37	-0.11	8.6	<2e-16
CVX	Energy	0.07	0.40	-1.24	28.7	<2e-16
AAPL	Technology	0.51	0.34	-0.43	9.5	<2e-16
MSFT	Technology	0.41	0.30	-0.48	14.8	<2e-16
PG	Consumer	0.22	0.23	0.03	16.2	<2e-16
MCD	Consumer	0.16	0.26	-0.44	37.5	<2e-16
PFE	Healthcare	0.16	0.28	0.16	8.5	<2e-16
JNJ	Healthcare	0.12	0.22	-0.11	11.7	<2e-16
BA	Industrials	-0.15	0.57	-0.40	15.7	<2e-16
DAL	Industrials	-0.07	0.54	-0.94	18.9	<2e-16

Is the normal model adequate? Table 2 gives a clear answer: no. Most stocks have negative skewness, so large losses are more common than equally large gains. Every stock has kurtosis far above the normal value of 3, with values from 8.5 to 37.5. This means that extreme returns occur much more often than a normal model predicts. The Jarque–Bera test rejects normality for all 12 stocks at any standard significance level; all reported p-values are below 2.2×10^{-16} . The QQ-plots in Figure 3 show the same result: the empirical points move away from the straight line in both tails, where risk measurement is most important.

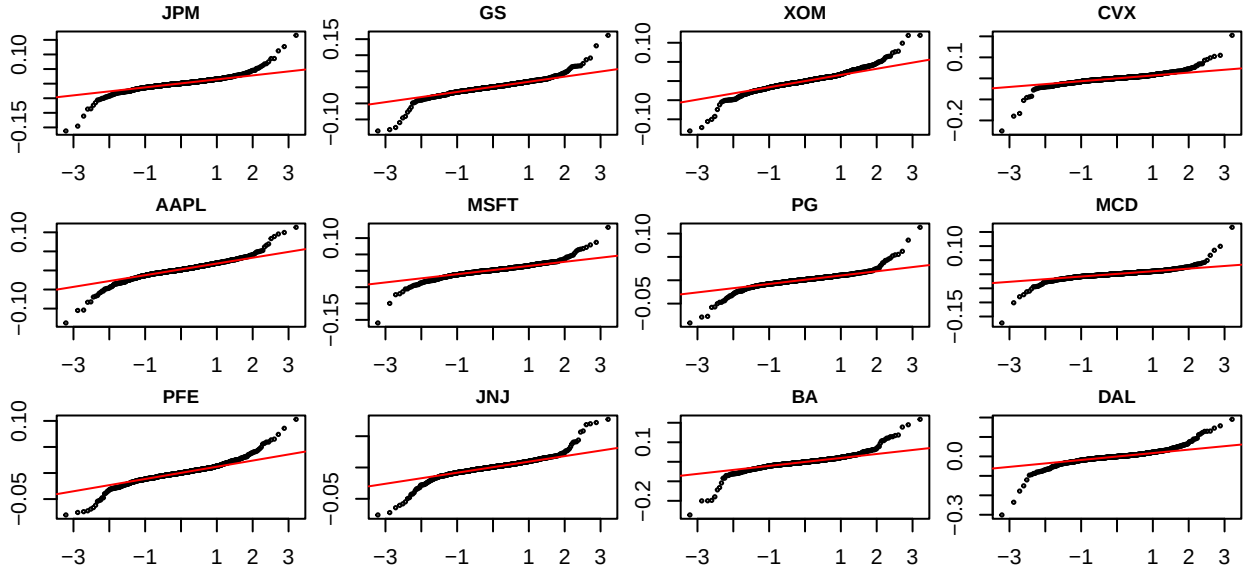


Figure 3: Normal QQ-plots of daily log returns. Deviations from the red line in the corners indicate heavy tails.

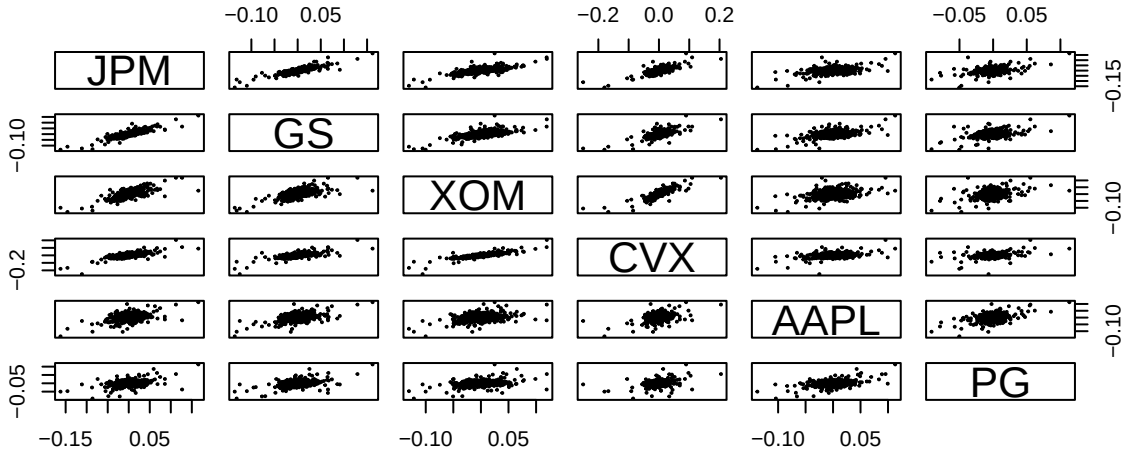


Figure 4: Pairwise daily log-return scatter plots for six selected stocks: financials (JPM, GS), energy (XOM, CVX), technology (AAPL) and consumer (PG).

What the scatter plots show. Figure 4 has two important features. Firstly, dependence is stronger within sectors than across sectors: JPM–GS and XOM–CVX form narrow clouds, with correlations of 0.86 and 0.85, while cross-sector pairs are more widely spread (for example, the AAPL–PG correlation is 0.47). Secondly, the most extreme observations appear together in the lower-left corners: during the crash days of March 2020, almost all stocks fell at the same time. Correlation measures average co-movement, but it does not fully describe this joint tail risk. This motivates the copula analysis in Section 5.

4. Portfolio Risk and Backtesting

Table 3: One-day VaR and ES of the portfolio, in % of portfolio value, estimated on 2019–2020.

	VaR 95%	VaR 99%	ES 95%	ES 99%
Empirical	2.53	6.00	4.57	8.43
Normal	2.83	4.04	3.57	4.63
Student-t	2.01	5.38	4.88	12.22

The portfolio. At the start of the sample, \$1,000 is invested in each stock, for a total of \$12,000. The number of shares remains fixed, so the portfolio is never rebalanced; its value reaches \$22,547 at the end of 2021. The daily loss is defined as $L_t = -(V_t/V_{t-1} - 1)$. Its distribution over 2019–2020 is modelled in three ways: empirically, using the historical quantiles of past losses; with a normal distribution; and with a heavy-tailed Student-t distribution fitted by maximum likelihood, which gives $\nu = 1.8$ degrees of freedom. This is an extremely heavy tail: for $\nu < 2$ the Student-t distribution has no finite variance, so far-tail estimates, especially the 99% ES, are sensitive to the fitted parameters and should be treated carefully.

Table 3 reports one-day VaR and ES at the 95% and 99% levels. In dollar terms, the 99% ES is \$1,012 for the empirical model, \$556 for the normal model and \$1,467 for the Student-t model, per \$12,000 invested. The three models are fairly close at 95%, but the differences are much larger at 99%, especially for ES — the average loss beyond VaR. The normal model gives the lowest far-tail estimates because its thin tails treat March-2020-style losses as extremely unlikely; the empirical and Student-t approaches give more weight to the observed extremes.

Table 4: Backtest on 2021: number of days the loss exceeded VaR, expected number, and the Kupiec unconditional-coverage test.

	Exceedances	Expected	LR stat	p-value
Empirical 95%	0	12.6	25.85	0.000
Normal 95%	0	12.6	25.85	0.000

	Exceedances	Expected	LR stat	p-value
Student-t 95%	3	12.6	10.97	0.001
Empirical 99%	0	2.5	5.07	0.024
Normal 99%	0	2.5	5.07	0.024
Student-t 99%	0	2.5	5.07	0.024

Backtesting on 2021. Each VaR estimate is applied, unchanged, to the 252 trading days of 2021. An *exceedance* occurs when the realised loss is larger than the VaR. A correct 95% VaR should be exceeded on about 5% of days, roughly 12.6 times, and a correct 99% VaR about 2.5 times. The Kupiec likelihood-ratio test in Table 4 checks whether the observed count is consistent with the target rate. At the 5% significance level, none of the six model-level combinations passes the test: every model produces fewer exceedances than expected. The Student-t 95% VaR is the closest to its target, with 3 exceedances against 12.6 expected, but it is still rejected. So the direct answer to *which approach held up out of sample* is: strictly, none of them.

The year 2021 was much calmer than the estimation window, which contained the worst crash in decades, so models estimated on crisis data are conservative in the test period — too few exceedances rather than too many. **What this says about model risk:** the same data and the same target can produce very different risk estimates, and backtest results depend on how similar the test period is to the estimation period. Risk measures should therefore be reported together with the model and estimation window that produced them, and thin-tailed models should be used carefully for far-tail measures such as the 99% ES.

5. Dependence and Copulas

Table 5: Copula families fitted to the pseudo-observations of the 12 stocks (2019–2020). Lower AIC indicates a better fit.

	Log-likelihood	Parameters	AIC
Gaussian	1885.1	66	-3638.3
Student-t	2176.2	67	-4218.3
Clayton	1044.6	1	-2087.2
Gumbel	913.1	1	-1824.2

A copula separates the *dependence structure* between the stocks from their individual return distributions. The 2019–2020 returns are transformed into pseudo-observations (their ranks scaled to the unit interval), and four copula families from the course are fitted: Gaussian, Student-t (both with an unstructured correlation matrix), Clayton and Gumbel. Table 5 gives an AIC-style comparison, based on the pseudo-likelihood of each fitted copula evaluated on the data. The **Student-t** copula gives the best fit. This is the expected outcome: the one-parameter Clayton and Gumbel copulas impose the same dependence on all 66 pairs of stocks, which is too restrictive for a portfolio from six sectors, while the Student-t copula uses a full correlation matrix and an extra tail parameter ($\nu = 3.8$ degrees of freedom).

Table 6: Probability (in %) that JPM and GS both fall below their u -quantile on the same day: empirical vs. simulated from the fitted copulas (100,000 draws, seed 317).

	Empirical	t copula	Gaussian	Independence
$u = 5\%$	2.58	3.22	2.82	0.25
$u = 1\%$	0.99	0.59	0.46	0.01

Tail dependence and simultaneous crashes. Tail dependence measures the probability that one stock has an extreme return *given* that another one does. The Gaussian copula has zero tail dependence for every pair: simultaneous large losses can still occur at a fixed threshold, but their conditional probability approaches zero as the threshold becomes more extreme. The fitted Student-t copula instead has positive tail dependence for all pairs, and stronger within sectors: for example $\lambda \approx 0.58$ for JPM–GS and $\lambda \approx 0.2$ for AAPL–PG. Table 6 tests whether the selected copula can reproduce simultaneous losses, comparing the probability that JPM and GS both fall below their u -quantile on the same day. At the 5% threshold both fitted copulas are close to the empirical probability of 2.58%: at this level the high correlation between the two stocks is the main driver, so the models are difficult to separate. At the 1% threshold the Student-t copula

gives 0.59%, compared with 0.46% for the Gaussian copula, 0.99% in the data and only 0.01% under independence. As the threshold becomes more extreme, the Gaussian probability falls much faster, so it may underestimate the chance of simultaneous crashes. One caveat: the empirical 1% value rests on only a few observations in a 504-day sample, so it is statistically noisy.

Why this matters for portfolio risk. Diversification protects a portfolio only when stocks do not crash together. Tail dependence means that during a crisis — when protection is most needed — co-movement increases and part of the diversification benefit disappears, exactly as March 2020 showed. A Gaussian dependence model may therefore underestimate the probability of several large losses on the same day, and hence underestimate portfolio VaR and ES at high confidence levels. Reliable risk numbers need to allow for both heavy-tailed returns and tail dependence.

6. Conclusion

Three main results emerge. Firstly, returns during the COVID-19 period were strongly non-normal: every stock has heavy tails, all normality tests are rejected, and many extreme days are linked to major news events in March and November 2020. Secondly, the choice of loss model matters most in the far tail, where it is hardest to test: the normal model gives a lower 99% ES than the empirical and Student-t models, and the 2021 backtest shows that risk estimates also depend on the estimation window (model risk). Thirdly, dependence changes under stress: the Student-t copula has the best AIC-style fit and, unlike the Gaussian copula, allows dependence to remain positive in increasingly extreme tails. The practical lesson is simple: risk models should allow for heavy tails and tail dependence, and their results should always be backtested.

Appendix: Full R Code

```
#I have done my project, using the rendering to pdf, LaTeX and this construction uses for it
knitr::opts_chunk$set(echo = FALSE, warning = FALSE, message = FALSE,
  fig.align = "center", fig.pos = "H",
  dev = c("pdf", "png"), dpi = 300)

library(quantmod)#download financial data, for ex. prices of stocks from Yahoo
library(moments)#compute skewness and kurtosis
library(MASS)#selects distributions, in our case, student's T, using fitdistr()
library(copula)#create and evaluate copula-models
library(xts)#works with time series, based on dates
library(knitr)#inserts code, tables, and graphs into R Markdown/PDF
set.seed(317)#just as our course ;)
tickers <- c("JPM", "GS", #financials
  "XOM", "CVX", #energy
  "AAPL", "MSFT", #technology
  "PG", "MCD", #consumer sector
  "PFE", "JNJ", #healthcare
  "BA", "DAL")#industrials and airlines
sectors <- rep(c("Financials", "Energy", "Technology", "Consumer",
  "Healthcare", "Industrials"), each = 2)#for each ticker its sector
getSymbols(tickers, src = "yahoo", from = "2019-01-01", to = "2022-01-01")
prices <- do.call(merge, lapply(tickers, function(t) Ad(get(t))))#common table of all prices
colnames(prices) <- tickers#rename columns to tickers names
n_miss <- sum(is.na(prices))#missing values before cleaning
if (n_miss > 0) {#impute only if gaps exist, in our case, no
  prices <- na.locf(prices)#carry last price forward
  prices <- na.omit(prices)#drop incomplete leading rows
}
simple_returns <- prices / lag(prices) - 1#compute simple returns
n_anom <- sum(abs(simple_returns) > 0.5, na.rm = TRUE)#big changes > 50% = possible errors
rets <- na.omit(diff(log(prices)))#daily log returns (drops the NA 1st row, as for 1st no previous)
rets_in <- rets["2019/2020"]#choose learning sample
rets_out <- rets["2021"]#backtesting samples
norm_p <- sweep(coredata(prices), 2, as.numeric(prices[, 1]), "/")#normalization of prices
matplot(index(prices), norm_p, type = "l", lty = 1, col = 1:12,
  xlab = "", ylab = "Price (start = 1)")
legend("topleft", legend = tickers, col = 1:12, lty = 1, cex = 0.55, ncol = 4)#1st graph of stock prices trend
abline(v = as.Date("2020-02-20"), lty = 2)#vertical line of the beginning of market crash due to COVID19
avg_ret <- xts(rowMeans(rets), order.by = index(rets))#avg returns of portfolio (all 12 stocks)
op <- par(mar = c(2, 4, 0.3, 0.5))#change margins for more accurate representation
plot(index(avg_ret), coredata(avg_ret), type = "h", col = "grey30",
  xlab = "", ylab = "Average log return")#2nd graph of avg log returns of portfolio
abline(h = 0, col = "red", lwd = 0.5)#add on graph red thin line at level 0
par(op)#return previous chart settings
av <- as.numeric(coredata(avg_ret))#take value of avg returns without dates
dts <- index(avg_ret)#save the corresponding dates
ord <- order(av)#work on plain vectors: xts re-sorts by date (in ascending order)
sel <- c(ord[1:5], rev(tail(ord, 5)))#5 worst days, then 5 best days
ext <- data.frame(
  Date = format(dts[sel], "%d %b %Y"),
  `Avg log return (%)` = round(100 * av[sel], 2),
  Type = rep(c("Worst", "Best"), each = 5), check.names = FALSE)#create table
kable(ext, caption = "Five worst and five best days, measured by the equal-weighted average log return of the 12 stocks.")#print table
jb_p <- apply(rets, 2, function(x) jarque.test(as.numeric(x))$p.value)#compute p-value of Jarque test for each column
stats <- data.frame#create table of stat. for each stock
Sector = sectors, add sector for each company
`Ann. mean` = round(colMeans(rets) * 252, 2), #compute annual avg log-returns
`Ann. vol.` = round(apply(rets, 2, sd) * sqrt(252), 2), #annual volatility, 252 - # of trading days in a year
Skewness = round(apply(rets, 2, skewness), 2), #skewness (negative - left tail bigger, positive - right one bigger)
Kurtosis = round(apply(rets, 2, kurtosis), 1), #usual for norm.dist. 3 is ok
`JB p-value` = format.pval(jb_p, digits = 2, eps = 2.2e-16), #format p-value
check.names = FALSE)#don't allow to change names
kable(stats, caption = "Summary statistics of daily log returns, 2019--2021")
```

```

(a normal distribution has skewness 0 and kurtosis 3.)"#display table
op <- par(mfrow = c(3, 4), mar = c(2, 2, 1.5, 0.5))#change a format for convenient visualization
for (t in tickers) {#starts a loop over all stock tickers
  qqnorm(as.numeric(rets[, t]), main = t, cex = 0.35, cex.main = 0.9, #3rd graph
    xlab = "", ylab = "")
  qqline(as.numeric(rets[, t]), col = "red")#adds a red reference line to the Q-Q plot
}
par(op)#again to previous settings
pairs(coredata(rets[, c("JPM", "GS", "XOM", "CVX", "AAPL", "PG")]),
  pch = 20, cex = 0.3)#4th graph
cors <- cor(coredata(rets))#calculates the full correlation matrix of all stock returns
shares <- 1000 / as.numeric(prices[1, ])$1000 per stock at the start
V <- xts(coredata(prices) %>% shares, order.by = index(prices))#total portfolio value on every date, as an xts time series
port_ret <- as.numeric(V / lag(V) - 1)[-1]#daily simple returns of the portfolio
loss <- xts(-port_ret, order.by = index(prices)[-1])#daily portfolio loss = negative of the portfolio return
loss_in <- as.numeric(loss["2019/2020"])#selects portfolio losses from 2019-2020 as the in-sample estimation data
loss_out <- as.numeric(loss["2021"])#selects portfolio losses from 2021 as the out-of-sample backtesting data
p_levels <- c(0.95, 0.99)#defines the two confidence levels used for risk measures: 95% and 99%
VaR_emp <- quantile(loss_in, p_levels)#calculates empirical VaR at the 95% and 99% levels using historical in-sample losses
ES_emp <- sapply(VaR_emp, function(v) mean(loss_in[loss_in >= v]))
#calculates empirical ES: the average loss on days when loss is at least VaR
mu <- mean(loss_in);
s <- sd(loss_in)#estimates the mean and standard deviation of portfolio losses for the normal model
VaR_nor <- mu + s * qnorm(p_levels)#calculates normal-theory VaR using the fitted mean and standard deviation
ES_nor <- mu + s * dnorm(qnorm(p_levels)) / (1 - p_levels)#calculates Expected Shortfall under the normal distribution assumption
m <- fitdistr(loss_in, "t")#fits Student-t distribution to the in-sample portfolio losses by maximum likelihood
n <- ft$estimate["n"]; sc <- ft$estimate["s"]; nu <- ft$estimate["df"]#extracts parameters of the Student-t distribution
qt_p <- qt(p_levels, df = nu)#calculate 95% and 99% quantiles
VaR_t <- m + sc * qt_p#calculates Student-t VaR using the estimated location, scale, and t-quantiles
ES_t <- m + sc * (dt(qt_p, nu) / (1 - p_levels)) * ((nu + qt_p^2) / (nu - 1))#calculates ES under the fitted Student-t distribution
risk <- rbind(Empirical = c(VaR_emp, ES_emp),
  Normal = c(VaR_nor, ES_nor),
  `Student-t` = c(VaR_t, ES_t))#combines the VaR and ES estimates from the three methods into one matrix
colnames(risk) <- c("VaR 95%", "VaR 99%", "ES 95%", "ES 99%")#assigns clear column names to the risk matrix
kable(risk * 100, 2)#displays the VaR and ES table in percentages, rounded to two decimals
caption = "One-day VaR and ES of the portfolio, in % of portfolio value, estimated on 2019-2020."
kupiec <- function(losses, VaR, p) {#defines a function for the Kupiec unconditional coverage test
  n <- length(losses); x <- sum(losses > VaR); phat <- x / n#counts the number of out-of-sample observations
  #and how many times the realised loss exceeded VaR, calculates the observed exceedance rate
  l10 <- x * log(p) + (n - x) * log(1 - p) #log-likelihood under H0
  l11 <- (if (x > 0) x * log(phat) else 0) + #log-likelihood at the MLE,
    (if (x < n) (n - x) * log(1 - phat) else 0)#robust to x = 0 or n
  LR <- -2 * (l10 - l11)#calculates the likelihood-ratio test statistic
  c(Exceedances = x, Expected = round(n * p, 1),
    `LR stat` = round(LR, 2), `p-value` = round(1 - pchisq(LR, 1), 3))
}#returns the counts, the LR statistic and its p-value
bt <- rbind(`Empirical 95%` = kupiec(loss_out, VaR_emp[1], 0.05),
  `Normal 95%` = kupiec(loss_out, VaR_nor[1], 0.05),
  `Student-t 95%` = kupiec(loss_out, VaR_t[1], 0.05),
  `Empirical 99%` = kupiec(loss_out, VaR_emp[2], 0.01),
  `Normal 99%` = kupiec(loss_out, VaR_nor[2], 0.01),
  `Student-t 99%` = kupiec(loss_out, VaR_t[2], 0.01))
#applies the Kupiec test to all three VaR models at the 95% and 99% levels and combines the results into one matrix
kable(bt, caption = "Backtest on 2021: number of days the loss exceeded VaR, expected number, and the Kupiec unconditional-coverage test.")
#display it
U <- pobs(as.matrix(rets_in)); d <- ncol(U)#converts the in-sample returns into pseudo-observations on the interval (0,1)
#and stores the number of variables, which is the copula dimension
fit_gauss <- fitCopula(normalCopula(dim = d, dispstr = "un"), U, method = "itau")
#fits a Gaussian copula to the pseudo-observations using inversion of Kendall's tau
fit_t <- fitCopula(tCopula(dim = d, dispstr = "un"), U, method = "itau.mpl")
#fits a Student-t copula using Kendall's tau for dependence parameters and maximum pseudo-likelihood for the degrees of freedom
fit_clay <- fitCopula(claytonCopula(dim = d), U, method = "mpl")#fits a Clayton copula by maximum pseudo-likelihood
fit_gum <- fitCopula(gumbelCopula(dim = d), U, method = "mpl")#fits a Gumbel copula by maximum pseudo-likelihood
ll <- c(Gaussian = sum(dCopula(U, fit_gauss@copula, log = TRUE)),
  `Student-t` = sum(dCopula(U, fit_t@copula, log = TRUE)),
  Clayton = sum(dCopula(U, fit_clay@copula, log = TRUE)),
  Gumbel = sum(dCopula(U, fit_gum@copula, log = TRUE)))#calculates the total log-likelihood for each fitted copula
npar <- c(d*(d-1)/2, d*(d-1)/2 + 1, 1, 1)#specifies the number of estimated parameters in each copula model
aic <- -2 * ll + 2 * npar#calculates the Akaike Information Criterion for each copula
kable(data.frame(`Log-likelihood` = round(ll, 1), Parameters = npar,
  AIC = round(aic, 1), check.names = FALSE),#creates and displays a comparison table of the fitted copula model
caption = "Copula families fitted to the pseudo-observations of the 12 stocks (2019-2020). Lower AIC indicates a better fit.")
nu_c <- coef(fit_t)[`df`]#extracts the estimated degrees of freedom from the fitted t-copula
rho_v <- coef(fit_t)[1:(d*(d-1)/2)]#extracts all pairwise dependence parameters from the fitted t-copula
lam <- -2 * pt(-sqrt((nu_c + 1) * (1 - rho_v) / (1 + rho_v)), df = nu_c + 1)
#calculates the pairwise tail-dependence coefficients of the t-copula
lam_M <- p2P(as.numeric(lam), d = d); dimnames(lam_M) <- list(tickers, tickers)
#converts the vector of pairwise tail-dependence coefficients into a symmetric matrix
#and assigns stock tickers as row and column names of the matrix
set.seed(317)#fixes the random-number generator for reproducibility
i <- which(tickers == "JPM"); j <- which(tickers == "GS")#finds the position of JPM/GS in the ticker vector
sim_t <- rCopula(1e5, fit_t@copula)#simulates 100,000 observations from the fitted t-copula
sim_g <- rCopula(1e5, fit_gauss@copula)#simulates 100,000 observations from the fitted Gaussian copula
jp <- function(M, u) mean(M[, i] < u & M[, j] < u)#joint tail probability
jc <- rbind(`u = 5%` = c(Empirical = jp(U, .05), `t copula` = jp(sim_t, .05),
  Gaussian = jp(sim_g, .05), Independence = .05^2),
  `u = 1%` = c(jp(U, .01), jp(sim_t, .01), jp(sim_g, .01), .01^2))
kable(round(100 * jc, 2),
caption = "Probability (in %) that JPM and GS both fall below their
u-quantile on the same day: empirical vs. simulated from the fitted
copulas (100,000 draws, seed 317).")

```

Appendix 1: Acknowledgement: The use of generative AI tools

I acknowledge the use of ChatGPT (GPT-5; OpenAI, <https://chatgpt.com/>) to translate individual words and short phrases from my native language into English, to proofread my own writing for grammar, spelling, and clarity, and to review my R code for possible syntax errors. The tool was not used to generate the analysis, interpretations, calculations, or results. All content was produced, checked, and verified by me, and I reviewed and implemented any code corrections myself.